RAJALAKSHMI ENGINEERING COLLEGE

Srikanth R 251801328-2025-AIDS S2

Started on	Thursday, 24 September 2026, 9:05 AM
State	Finished
Completed on	Thursday, 24 September 2026, 9:25 AM
Time taken	20 mins 11 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>

int main(){
    char s1[100],s2[100];
    scanf("%s",s1);
    scanf("%s",s2);
    int n=strlen(s1);
    int m=strlen(s2);
    int dp[n+1][m+1];

    for(int i=0;i<=n;i++){
        dp[i][0]=0;
    }
    for(int j=0;j<=m;j++){
        dp[0][j]=0;
    }

    for(int i=1;i<=n;i++){
        for(int j=1;j<=m;j++){
            if(s1[i-1]==s2[j-1]){
                dp[i][j]=dp[i-1][j-1]+1;
            }
            else{
                if(dp[i-1][j]>dp[i][j-1]){
                    dp[i][j]=dp[i-1][j];
                }
                else{
                    dp[i][j]=dp[i][j-1];
                }
            }
        }
    }
    printf("%d",dp[n][m]);
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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